

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO3:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm

converges faster based on the observed results.
(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.
(5 Marks)

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.
(5 Marks)

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.
(5 Marks)

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.
(5 Marks)

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.
(5 Marks)

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.
(5 Marks)

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.
(5 Marks)

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

CO3 AT3 – DATA INTERPRETATION: REINFORCEMENT LEARNING FOR ENVIRONMENT AND AGENT

Student Name	:	Shaik Ahamed Ali
Registration Number	:	192425106
Course Code & Name	:	MLA0302 - Reinforcement Learning
Date of Submission	:	13-08-2026

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. (5 Marks)

Aim

To interpret the trend of increasing episodic reward values and to assess what this trend reveals about the agent's learning performance and the improvement of its policy over time.

Given

An RL agent trained over multiple episodes, with reward values observed to increase gradually from 10 to 45.

Understanding of Data

The data shows a clear monotonic upward trend in cumulative reward, rising from an initial value of 10 to 45 across successive training episodes. This steady, gradual increase—rather than a flat or erratic pattern—indicates that the reward signal is being consistently translated into improved agent behaviour over time, and there is no evidence of divergence or instability in the observed values.

Analysis

In reinforcement learning, the reward obtained by an agent is a direct measure of how well its current policy is performing in the environment. A gradual rise from 10 to 45 suggests that with

every batch of episodes, the agent's value-function estimates are becoming more accurate and its action-selection is shifting away from random or sub-optimal choices toward actions that yield higher long-term returns. The smoothness of the increase (as opposed to sudden jumps or oscillations) further implies that the learning rate and exploration schedule are well balanced, allowing stable gradient/value updates rather than overshooting.

Application of Concepts

This pattern is consistent with the theoretical expectation from Dynamic Programming-based methods such as Policy Iteration and Value Iteration, where repeated application of the Bellman update causes the value function—and consequently the reward obtained by the derived policy—to converge monotonically toward the optimal value function V^* . The trend can therefore be interpreted through the lens of policy improvement theorem, which guarantees that a greedy policy with respect to an improving value function is itself non-decreasing in performance.

Conclusion

The rising reward trend is strong evidence of successful learning and progressive policy improvement. It indicates that the agent is converging toward, or approaching, an optimal policy, though final confirmation of true optimality would require observing the reward curve plateau at a stable maximum over further episodes.

Clarity

In summary: Reward trend = 10 $\rightarrow$ 45 (monotonic increase); Interpretation = consistent policy improvement via repeated value updates; Key terms used = value function, Bellman update, policy improvement theorem, convergence; Overall verdict = positive learning trend, likely approaching optimal policy.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. (5 Marks)

Aim

To apply the concept of exploration versus exploitation in order to interpret why cumulative reward varies with epsilon and to identify the epsilon value that yields the best performance.

Given

An epsilon-greedy agent tested at $\epsilon = 0.1$, 0.3 and 0.5 , with cumulative reward highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$.

Understanding of Data

The data indicates an inverse relationship between epsilon and cumulative reward: as epsilon increases from 0.1 to 0.5 , the cumulative reward decreases, with $\epsilon = 0.1$ producing the best performance and $\epsilon = 0.5$ the worst.

Analysis

In an epsilon-greedy strategy, the agent selects the currently best-known action (exploitation) with probability $(1 - \epsilon)$ and a random action (exploration) with probability ϵ . A low epsilon such as 0.1 means the agent exploits its learned knowledge 90% of the time, allowing it to consistently capitalise on actions already known to yield high reward, while still reserving a small proportion of trials to discover potentially better actions. As epsilon rises to 0.5 , the agent behaves almost as randomly as a uniform policy half the time, which severely limits its ability to exploit good actions consistently, thereby lowering the average and cumulative reward.

Application of Concepts

This directly reflects the exploration–exploitation trade-off central to reinforcement learning. Excessive exploration (high epsilon) is useful only in the early stages of learning when the agent's estimates of action values are unreliable; once reasonable estimates are formed, continued high exploration wastes opportunities to exploit already-known good actions. The results align with the standard theoretical prediction that very high epsilon values increase regret because the agent forgoes exploitation too frequently.

Conclusion

$\epsilon = 0.1$ provides the best performance because it maintains sufficient exploitation of the learned policy while allowing minimal exploration to refine action-value estimates. $\epsilon = 0.5$ is sub-optimal, as excessive randomness prevents the agent from consistently leveraging its

acquired knowledge, confirming that a small epsilon is preferable once the agent has gathered adequate experience.

Clarity

In summary: Epsilon values tested = 0.1, 0.3, 0.5; Trend = cumulative reward decreases as epsilon increases; Key terms used = exploration, exploitation, epsilon-greedy, regret; Overall verdict = epsilon = 0.1 is the best-performing configuration.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)

Aim

To identify the performance difference between Q-Learning and SARSA based on their average rewards and to determine which algorithm converges faster.

Given

Average rewards for Q-Learning and SARSA recorded at 100, 200 and 300 episodes, with Q-Learning consistently showing slightly higher reward values than SARSA.

Understanding of Data

Across all three checkpoints (100, 200 and 300 episodes), Q-Learning consistently records a marginally higher average reward than SARSA, and this gap persists throughout training rather than narrowing or reversing.

Analysis

The consistent edge of Q-Learning can be explained by the fundamental difference in how the two algorithms update their value estimates. Q-Learning is an off-policy method that updates its Q-values using the maximum reward achievable from the next state ($\max_a Q(s',a)$), regardless of the action actually taken by the current exploratory policy. SARSA, in contrast, is an on-policy method that updates using the reward of the action actually selected under the current epsilon-greedy policy. Because Q-Learning always bootstraps toward the greedy/optimal action,

it tends to learn a more aggressive, higher-reward-seeking policy faster, whereas SARSA's updates are tempered by the exploratory actions it actually takes, making it more conservative and often slower to reach high-reward regions.

Application of Concepts

This behaviour matches the well-established theoretical distinction between off-policy and on-policy control in Dynamic-Programming-based RL: Q-Learning directly approximates the optimal action-value function Q^* , while SARSA converges to the value function of the policy actually being followed (including exploration), which is typically slightly lower in expected return, especially in environments with risk or penalty states.

Conclusion

Q-Learning converges faster and achieves marginally higher rewards because its off-policy, max-based update aggressively pursues the optimal policy irrespective of the exploration strategy, whereas SARSA's on-policy nature makes it more cautious. For tasks prioritising speed of convergence to high reward, Q-Learning is the preferable algorithm, though SARSA may still be favoured in safety-critical settings due to its more conservative learning.

Clarity

In summary: Algorithms compared = Q-Learning vs SARSA; Checkpoints = 100, 200, 300 episodes; Trend = Q-Learning consistently ahead; Key terms used = off-policy, on-policy, bootstrapping, optimal action-value function Q^* ; Overall verdict = Q-Learning converges faster than SARSA.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. (5 Marks)

Aim

To discuss how the discount factor affects long-term reward optimisation and to justify which discount factor value is most suitable for achieving better learning performance.

Given

An RL agent trained with discount factor values 0.2, 0.5 and 0.9, with average reward observed to increase as the discount factor increases.

Understanding of Data

The data shows a positive correlation between the discount factor (gamma) and average reward: as gamma increases from 0.2 to 0.9, the average reward obtained by the agent also increases.

Analysis

The discount factor gamma determines the weight the agent places on future rewards relative to immediate ones, as defined by the return $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$. A low gamma such as 0.2 makes the agent almost myopic, causing it to heavily discount future consequences and optimise only for immediate reward, which is sub-optimal in tasks where long-term strategy matters. As gamma approaches 0.9, the agent increasingly values future rewards nearly as much as immediate ones, enabling it to plan further ahead and choose actions that may yield lower immediate reward but substantially higher cumulative reward over the episode.

Application of Concepts

This pattern is consistent with Dynamic Programming principles underlying the Bellman equation, where the discount factor directly governs the horizon of optimisation considered in the value function $V(s) = \sum \gamma^t R_{t+1}$. Higher gamma values bring the computed policy closer to a truly optimal long-term policy, provided the environment is not purely episodic/short-horizon, in which case excessively high gamma could also introduce higher variance in value estimates.

Conclusion

A discount factor of 0.9 is most suitable in this scenario, as it allows the agent to optimise for long-term cumulative reward rather than short-sighted immediate gains, resulting in the best observed performance. This confirms that higher discount factors generally support better learning performance in tasks requiring long-horizon planning.

Clarity

In summary: Discount factors tested = 0.2, 0.5, 0.9; Trend = average reward increases with gamma; Key terms used = discount factor, return, Bellman equation, long-horizon planning; Overall verdict = gamma = 0.9 yields the best learning performance.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. (5 Marks)

Aim

To discuss the improvement in the agent's success rate and to interpret, with reasoning based on the data trend, whether the agent has learned an optimal policy.

Given

Success rate of an RL agent recorded as 45% at 50 episodes and 92% at 200 episodes.

Understanding of Data

The success rate more than doubles, rising from 45% at 50 episodes to 92% at 200 episodes, indicating a strong and consistent upward trend in the agent's task-completion capability as training progresses.

Analysis

Such a sharp rise in success rate demonstrates that the agent's policy is becoming increasingly reliable at achieving the desired goal state. Early in training (50 episodes), the agent's policy is largely under-developed due to limited experience and higher reliance on exploratory/random actions, resulting in a low success rate. As training progresses to 200 episodes, repeated policy evaluation and improvement cycles allow the value estimates to stabilise, and the agent increasingly selects actions that reliably lead to successful outcomes, reflected in the 92% success rate.

Application of Concepts

This closely mirrors the Generalized Policy Iteration (GPI) framework, where alternating steps of policy evaluation and policy improvement progressively push the policy toward optimality. A success rate approaching (but not reaching) 100% suggests the agent is near-optimal but may not yet be fully optimal, since a small fraction of episodes still result in failure, possibly due to residual exploration, environmental stochasticity, or insufficient training episodes for complete convergence.

Conclusion

The improvement in success rate strongly indicates effective learning and substantial policy improvement, but a 92% success rate—while excellent—does not by itself confirm that a fully optimal policy has been learned. Continued training and monitoring for a plateau at or very near 100% would be needed to conclusively establish optimality.

Clarity

In summary: Success rate = 45% (50 episodes) → 92% (200 episodes); Trend = strong upward improvement; Key terms used = policy evaluation, policy improvement, Generalized Policy Iteration; Overall verdict = near-optimal policy, not yet conclusively optimal.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. (5 Marks)

Aim

To analyze how the learning rate influences convergence in the CartPole environment and to justify which learning rate value offers the best trade-off between learning speed and stability.

Given

An RL agent trained in the CartPole environment with learning rates 0.01, 0.05 and 0.1, achieving the highest stability at 0.05.

Understanding of Data

Among the three tested learning rates (0.01, 0.05 and 0.1), the agent demonstrates the highest stability when the learning rate is set to 0.05, while the extreme values on either side (0.01 and 0.1) result in comparatively lower stability.

Analysis

The learning rate (α) controls the magnitude of the update applied to value or policy parameters after each experience. A very small learning rate of 0.01 causes updates to be extremely gradual, meaning the agent learns very slowly and may not sufficiently adjust its

policy within the CartPole environment's episode limits, leading to slow, sometimes unstable improvement due to insufficient correction of errors. Conversely, a high learning rate of 0.1 causes excessively large updates to the value/policy estimates, which can overshoot optimal parameter values, causing oscillations and instability in the balancing behaviour required for CartPole. The intermediate value of 0.05 strikes a balance, allowing sufficiently large updates to learn quickly while remaining small enough to avoid destabilising the policy.

Application of Concepts

This reflects the well-known trade-off in gradient-based and temporal-difference learning: convergence speed and stability are inversely sensitive to the learning rate, and there exists an optimal intermediate range (illustrated here by 0.05) where the update step size is neither too conservative nor too aggressive, consistent with convergence guarantees for stochastic approximation methods that require the learning rate to be neither too large nor too small.

Conclusion

A learning rate of 0.05 provides the best trade-off between convergence speed and policy stability, as it avoids the slow learning associated with 0.01 and the oscillatory, unstable behaviour associated with 0.1, making it the most suitable choice for training a stable CartPole balancing agent.

Clarity

In summary: Learning rates tested = 0.01, 0.05, 0.1; Trend = highest stability at 0.05; Key terms used = learning rate (alpha), convergence, oscillation, stochastic approximation; Overall verdict = 0.05 gives the best speed–stability trade-off.

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. (5 Marks)

Aim

To compare epsilon-greedy, softmax, and random exploration strategies and to defend why epsilon-greedy achieves better performance in this scenario.

Given

A Q-Learning agent tested with three exploration strategies—epsilon-greedy, softmax, and random exploration—with cumulative reward highest under epsilon-greedy.

Understanding of Data

Among the three exploration strategies tested—epsilon-greedy, softmax, and random exploration—the epsilon-greedy strategy yields the highest cumulative reward, indicating it is the most effective exploration mechanism in this scenario.

Analysis

Random exploration selects actions with no regard to their estimated value, meaning the agent wastes a large number of trials on clearly sub-optimal actions, which severely limits cumulative reward. Softmax exploration improves upon this by weighting action-selection probabilities according to their estimated Q-values, but it can still assign non-trivial probability to poor actions when the temperature parameter is not finely tuned, and its performance is highly sensitive to that parameter. Epsilon-greedy, by contrast, exploits the best-known action with high probability ($1 - \epsilon$) while reserving only a small, controlled probability for exploration, giving it a simple but effective balance that consistently favours high-value actions once reasonable estimates are formed.

Application of Concepts

This outcome is consistent with the exploration–exploitation trade-off discussed in Q-Learning theory: strategies that exploit current knowledge more consistently, while retaining minimal but non-zero exploration, tend to accumulate higher rewards than strategies that explore too broadly (random) or that require careful hyperparameter tuning to perform well (softmax). Epsilon-greedy's simplicity and robustness to parameter choice make it particularly effective in moderately complex environments.

Conclusion

Epsilon-greedy achieves the best performance because it offers a favourable and easily controlled balance between exploiting the current best policy and exploring alternative actions, avoiding the excessive randomness of pure random exploration and the tuning sensitivity of softmax exploration, thereby maximising cumulative reward in this scenario.

Clarity

In summary: Strategies compared = epsilon-greedy, softmax, random; Result = epsilon-greedy achieves highest cumulative reward; Key terms used = exploitation, exploration, temperature parameter, robustness; Overall verdict = epsilon-greedy is the most effective and reliable strategy here.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. (5 Marks)

Aim

To interpret the reduction in the number of steps taken to reach the goal and to evaluate whether this reduction indicates convergence to an optimal policy.

Given

An RL agent trained in a maze environment, initially taking 50 steps to reach the goal and reducing to 15 steps after 300 episodes.

Understanding of Data

The number of steps required to reach the goal decreases substantially from 50 steps at the start of training to just 15 steps after 300 episodes, showing a clear improvement in the efficiency of the agent's path-finding behaviour.

Analysis

In a maze environment, the number of steps taken to reach the goal is a direct proxy for policy quality: a poor or undeveloped policy leads the agent to wander, revisit states, or take unnecessary detours, while a well-learned policy guides the agent along a near-shortest path. The steady reduction from 50 to 15 steps indicates that, through repeated episodes of trial and error combined with value-based updates (e.g., via Q-Learning or SARSA), the agent has progressively eliminated inefficient actions and learned a more direct route to the goal.

Application of Concepts

This is consistent with the Dynamic Programming notion of policy improvement, where each iteration of value updates and greedy action-selection should not increase (and typically decreases) the expected number of steps to the goal, since shorter paths correspond to higher discounted cumulative reward (especially under a step-penalty reward structure). If 15 steps corresponds to, or is close to, the true shortest path length in the maze, this would be strong evidence that the agent has reached, or nearly reached, an optimal policy.

Conclusion

The reduction in steps strongly suggests convergence toward an optimal, or near-optimal, policy, as the agent is consistently finding shorter paths to the goal. Full confirmation of optimality, however, would require comparing the 15-step result against the known shortest path length for the maze and verifying that further training does not reduce the step count any further.

Clarity

In summary: Steps to goal = 50 (initial) → 15 (after 300 episodes); Trend = strong reduction, indicating efficiency gain; Key terms used = policy improvement, shortest path, step-penalty reward; Overall verdict = near-optimal policy achieved, pending shortest-path verification.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. (5 Marks)

Aim

To discuss the role of the discount factor in balancing immediate versus future rewards and to justify why a discount factor of 0.95 is most effective for the SARSA agent.

Given

A SARSA agent trained with discount factors 0.4, 0.7 and 0.95, achieving the highest long-term reward with 0.95.

Understanding of Data

The data shows that as the discount factor increases from 0.4 to 0.95, the long-term reward achieved by the SARSA agent also increases, with 0.95 producing the best overall performance.

Analysis

The discount factor γ in the return formula $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$ controls how strongly future rewards influence the agent's current decisions. A low discount factor of 0.4 causes the agent to heavily discount rewards received several steps into the future, effectively making it near-sighted and prone to choosing actions that maximise only immediate gains, even if this leads to lower total reward over the episode. As γ increases to 0.95, the agent gives future rewards almost equal weight to immediate rewards, encouraging it to adopt strategies that may sacrifice small short-term rewards in favour of substantially larger long-term returns.

Application of Concepts

Because SARSA is an on-policy algorithm that learns the value of the policy it actually follows (including exploratory actions), a higher discount factor allows it to more accurately account for the long-run consequences of its exploratory behaviour, aligning its learned value function more closely with genuine long-term outcomes. This matches the Dynamic Programming principle that the value function $V(s)$ under the Bellman equation becomes more forward-looking as γ approaches 1, provided the environment is not purely short-horizon.

Conclusion

A discount factor of 0.95 is most effective because it enables the SARSA agent to prioritise long-term cumulative reward over short-sighted immediate gains, resulting in superior overall performance. This confirms that in tasks requiring extended planning, higher discount factors close to 1 are generally preferable.

Clarity

In summary: Discount factors tested = 0.4, 0.7, 0.95; Trend = long-term reward increases with γ ; Key terms used = discount factor, on-policy learning, return, forward-looking value function; Overall verdict = $\gamma = 0.95$ is the most effective setting.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. (5 Marks)

Aim

To examine the reward trend in a stochastic environment and to defend how the observed stability indicates policy improvement and robustness of learning.

Given

An RL agent trained in a stochastic environment, showing fluctuating rewards in early episodes and stabilising after 500 episodes.

Understanding of Data

The reward trend shows high fluctuation during the early episodes of training, followed by a clear stabilisation once the agent has completed approximately 500 episodes, after which reward values remain relatively consistent.

Analysis

The initial fluctuation is expected in a stochastic environment, where randomness in state transitions and/or rewards, combined with the agent's own exploratory behaviour and immature value estimates, causes high variance in episode-to-episode performance. As training progresses, the agent accumulates sufficient experience to form more accurate estimates of state and action values, and its policy becomes less reliant on exploratory, uncertain actions. The stabilisation after 500 episodes indicates that the agent's policy has converged to a consistent behaviour that reliably manages the inherent randomness of the environment, rather than being thrown off by it.

Application of Concepts

This trend reflects the theoretical expectation that, under adequate exploration and appropriately decaying learning parameters, value-based methods converge toward a stable, near-optimal value function even in stochastic settings, as guaranteed under standard stochastic approximation conditions. Stability in reward, rather than continued high variance, is a key empirical indicator that the policy is robust—capable of achieving consistent outcomes despite environmental randomness—rather than merely having achieved a single lucky high-reward episode.

Conclusion

The transition from fluctuating to stable rewards is strong evidence of both policy improvement and learning robustness, as it shows the agent has developed a policy that performs reliably under stochastic conditions rather than one that is sensitive to random variation. This stability is a meaningful indicator that the agent's learning process has successfully converged.

Clarity

In summary: Reward trend = fluctuating (early episodes) → stable (after 500 episodes); Interpretation = variance reduction indicates convergence; Key terms used = stochastic environment, variance, convergence, robustness; Overall verdict = stable rewards confirm a robust, well-converged policy.